



AI Playbook and Toolkit for Public Health Departments

Causal Inference and AI in Public Health Decision-Making

ANL 340

AI models often predict what is likely to happen, but public health leaders frequently need to know what action will change an outcome. This module helps learners distinguish prediction from causal inference so AI outputs are not misused as evidence that a program, policy, or intervention caused an observed effect. Learners examine confounding, bias, comparison groups, counterfactual reasoning, directed acyclic graphs, and quasi-experimental designs at a practical level. The purpose is not to turn every learner into a causal methods expert. The purpose is to help public health teams recognize when causal claims require stronger evidence than a predictive model can provide.

Level	intermediate/practitioner
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-modeling, epidemiology, policy, governance-security

Learning Objectives

- Distinguish prediction from causal inference in public health AI.
- Explain why correlation and prediction do not establish causation.
- Identify confounding, selection bias, collider bias, and temporal ambiguity in public health data.
- Recognize when a decision requires causal evidence rather than risk prediction alone.
- Identify appropriate evidence sources for evaluating interventions or policy effects.
- Create a causal question framing worksheet for an AI-supported decision.

Module Overview

AI models often predict what is likely to happen, but public health leaders frequently need to know what action will change an outcome. This module helps learners distinguish prediction from causal inference so AI outputs are not misused as evidence that a program, policy, or intervention caused an observed effect.

Learners examine confounding, bias, comparison groups, counterfactual reasoning, directed acyclic graphs, and quasi-experimental designs at a practical level. The purpose is not to turn every learner into a causal methods expert. The purpose is to help public health teams recognize when causal claims require stronger evidence than a predictive model can provide.

Definitions

- **Prediction:** An estimate of what is likely to happen based on observed patterns.
- **Causal inference:** The process of estimating whether and how an exposure, policy, program, or intervention changes an outcome.
- **Confounding:** A distortion that occurs when another factor is related to both the exposure and the outcome.
- **Counterfactual:** The outcome that would have occurred under a different condition or intervention.
- **Quasi-experimental design:** An evaluation approach that estimates causal effects without random assignment.

Why This Topic Matters for Public Health AI

AI can make weak evidence appear stronger than it is. A model may predict which communities have higher hospitalization risk, but that does not prove which intervention will reduce hospitalization. A model may find that people who receive a service have better outcomes, but that pattern may reflect who was able to access the service rather than the effect of the service itself.

Public health decisions often involve resource allocation, program design, policy evaluation, and equity tradeoffs. These decisions require clarity about whether the agency is asking a prediction question, an explanation question, or an intervention question. Causal inference helps agencies avoid treating AI-generated associations as proof of program effectiveness.

Public Health Example

A local health department uses an AI model to predict which neighborhoods are likely to experience low vaccination coverage. The model may help target outreach, but it does not prove why coverage is low or which outreach strategy will work. If leaders want to know whether mobile clinics, reminder messages, school partnerships, or community health workers caused an improvement, they need an evaluation design that supports causal inference.

This distinction prevents overclaiming. The AI model can support prioritization and hypothesis generation, while causal methods can support evaluation of the chosen intervention. Both are valuable, but they answer different questions.

Technical and Operational Deep Dive

Causal inference begins with a clearly framed question. A useful structure is: among which population, what is the intervention or exposure, what is the comparison, what outcome is measured, and over what time period? Without this structure, teams may ask a model to answer a question that the available data cannot support.

Confounding is one of the most important threats. For example, a program may appear to improve outcomes because participants were already more connected to care. Selection bias may occur when data include only people who interacted with a system. Temporal ambiguity may occur when it is unclear whether the exposure occurred before the outcome. AI can detect patterns in these data, but it does not automatically solve the causal design problem.

Methods such as matching, regression adjustment, difference-in-differences, interrupted time series, regression discontinuity, and instrumental variables may be useful when randomization is not possible. These methods require assumptions that should be documented and reviewed. Public health staff should know when to consult epidemiologists, statisticians, or evaluators before making causal claims.

Risks, Failure Modes, and Guardrails

The main failure mode is causal overreach. A dashboard or AI report may state that a program reduced disease, improved access, or changed behavior when the evidence only shows association. A guardrail is to require causal language review before results are used in reports, funding decisions, policy recommendations, or public communications.

Another risk is inequitable inference. Data may be more complete for some groups than others, and interventions may reach groups differently. A guardrail is to examine whether assumptions hold across populations and whether subgroup effects or access barriers are hidden by aggregate results.

Application to AI-Supported Workflows

Predictive models can identify risk, but causal inference is needed to decide what action is likely to change that risk. Generative AI may help draft evaluation questions or summarize literature, but humans must verify methods and assumptions. AI-supported decision-making should label outputs clearly as prediction, association, causal evidence, or hypothesis.

Reflection Questions

What AI outputs in your agency could be mistaken for causal evidence?

Which decisions require knowing what intervention will change an outcome?

What confounders are likely in your program area?

What comparison group could support evaluation?

Who should review causal claims before public release?

Practical Exercise

Create a causal question framing worksheet for an AI-supported decision. Identify the population, intervention or exposure, comparison, outcome, time period, likely confounders, data limitations, and what evidence would be

needed before making a causal claim.

Expected Artifact or Evidence

Causal question framing worksheet

List of assumptions and likely confounders

Recommended evaluation design or expert consultation need

Plain-language statement distinguishing prediction from causation

Expected Assignment

- Causal question framing worksheet
- List of assumptions and likely confounders
- Recommended evaluation design or expert consultation need
- Plain-language statement distinguishing prediction from causation

Knowledge Check

- 1. Which statement best distinguishes prediction from causal inference?
- 2. What is the best reason to assign an accountability owner?
- 3. When should an AI-supported workflow be escalated for review?
- 4. Which practice best supports public accountability?
- 5. What should happen when data sources, policy, vendor configuration, or workflow use changes?

References and Resources

- CDC Public Health Data Strategy and Data Modernization resources: <https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
- NIST Artificial Intelligence Risk Management Framework and NIST Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- WHO Ethics and Governance of Artificial Intelligence for Health guidance: <https://www.who.int/publications/i/item/9789240029200>
- OWASP Top 10 for Large Language Model Applications: <https://owasp.org/www-project-top-10-for-large-language-model-applications/>
- ONC health IT, interoperability, and information blocking resources: <https://www.healthit.gov/>
- Agency policies for privacy, public records, cybersecurity, procurement, civil rights, accessibility, language access, and emergency communications